

MODULE 5: VALUE ANALYSIS AND COST CONTROL

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Value Analysis (VA)

Value Analysis (VA) is a systematic approach to improving the value of a product or service by assessing its functions, costs, and performance. The goal is to reduce costs while maintaining or enhancing quality and functionality. **Terms Used in Value Analysis**

1. **Function:** The specific purpose or role that a product or service serves. Functions can be classified as basic (essential for the product's existence) or secondary (enhancements that add value).
2. **Cost:** The total expenses incurred in producing a product or service, including material, labour, overhead, and any additional expenses.
3. **Value:** The ratio of function to cost. A product has high value when its function is significant relative to its cost.
4. **Value Engineering (VE):** A subset of VA focused specifically on design and engineering aspects to improve product value.
5. **Functional Analysis:** The process of evaluating the functions of a product or service to determine their necessity and cost-effectiveness.

Process of Value Analysis

1. **Information Gathering:** Collect data on the product, including specifications, costs, and performance metrics.
2. **Function Analysis:** Identify and classify the functions of the product, distinguishing between essential and non-essential functions.
3. **Cost Assessment:** Analyse the costs associated with each function to determine which functions may be unnecessary or over-engineered.
4. **Idea Generation:** Brainstorm alternative solutions or improvements that could enhance functionality while reducing costs.

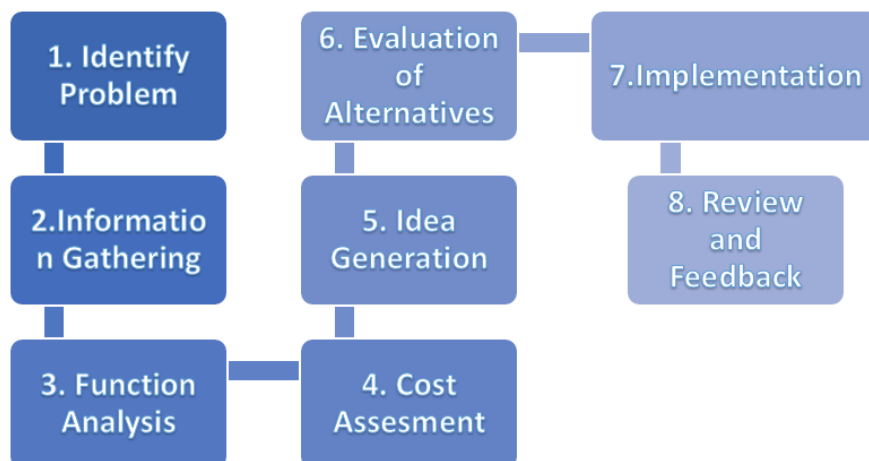
5. **Evaluation:** Assess the feasibility, practicality, and potential savings of proposed changes or alternatives.
6. **Implementation:** Develop an action plan to implement the approved changes, including timelines and responsible parties.
7. **Review and Feedback:** Monitor the outcomes of the implemented changes to ensure that desired improvements in value and cost are achieved.

Importance of Value Analysis

1. **Cost Reduction:** Identifies opportunities to eliminate unnecessary costs, contributing to overall savings.
2. **Enhanced Product Quality:** Encourages continuous improvement in product design and functionality, leading to higher quality.
3. **Increased Competitiveness:** Helps organizations offer better value propositions to customers, enhancing market positioning.
4. **Customer Satisfaction:** Focuses on improving product features that matter most to customers, leading to increased satisfaction.
5. **Innovation:** Encourages creative thinking and innovation by exploring alternative designs and materials.
6. **Resource Optimization:** Ensures efficient use of materials and labour, leading to more sustainable production practices.

Value Analysis (VA) Flow Diagram

A **Value Analysis Flow Diagram** visually represents the steps involved in the value analysis process. Here's a description of the key components typically included in the diagram:



1. **Identify the Problem:** Define the focus area for the value analysis, such as a specific product, service, or process that needs evaluation.
2. **Gather Information:** Collect relevant data, including specifications, costs, performance metrics, and customer feedback to understand the current situation.
3. **Analyse Functions:** Identify and categorize the essential and non-essential functions of the product or service to determine what adds value.
4. **Cost Assessment:** Analyse the costs associated with each function to understand their contribution to overall expenses.
5. **Generate Ideas:** Brainstorm potential alternatives or improvements that could enhance functionality while reducing costs.

6. **Evaluate Alternatives:** Assess the feasibility and potential impact of the proposed alternatives on cost and functionality.
7. **(Implementation):** Execute the selected ideas and integrate changes into the production or service process.
8. **Review and Feedback:** Monitor the impact of the changes, gather feedback, and assess whether the desired improvements in value and cost have been achieved.

DARSIRI Method of Value Analysis

The **DARSIRI** method is a structured approach in Value Analysis that consists of seven key steps, each corresponding to a specific phase in the VA process:

1. **D – Define:** Clearly define the problem or objective of the VA study. What is the goal of the analysis? **Outcome:** Establish a clear understanding of the product or service and its intended function.
2. **A – Analyse:** Analyse the product's current design and functionality to identify areas of improvement. **Outcome:** Break down the product or service into its components to understand the cost and value of each function.
3. **R – Research:** Investigate alternative methods, materials, or designs that could improve value. **Outcome:** Identify new opportunities for cost reduction or improved functionality.
4. **S – Speculate:** Brainstorm and generate creative ideas for enhancing the product's value. **Outcome:** Develop a range of potential solutions, with no restrictions on feasibility at this stage.
5. **I – Innovate:** **Purpose:** Evaluate the ideas generated during the speculation phase and select the most promising ones. **Outcome:** Refine and develop innovative solutions that offer the best value for money.
6. **R – Recommend:** Propose the best solution(s) based on the evaluation process. **Outcome:** Develop a detailed plan for implementing the recommended solution, including cost-benefit analysis.
7. **I – Implement:** Put the recommended solution into action. **Outcome:** Execute the changes and monitor their effectiveness to ensure that the desired improvements in value and cost savings are achieved.

Case Study on VA

Case Study 1: Automotive Industry – Redesigning a Car Door

Background: A major automobile manufacturer faced rising costs in its vehicle production, particularly concerning the weight and cost of car doors. The original design used heavy materials, leading to higher manufacturing costs and reduced fuel efficiency for the vehicles.

Value Analysis Process:

1. **Identify the Problem:** High production costs and weight of car doors affecting vehicle performance.
2. **Gather Information:** Collected data on the current design, materials used, manufacturing processes, and associated costs.
3. **Analyse Functions:** Determined the essential functions of the car door: protection, insulation, aesthetics, and access to the vehicle.

4. **Assess Costs:** Analysed the cost of materials, labour, and manufacturing processes related to each function of the car door.
5. **Generate Ideas:** Brainstormed alternative materials (e.g., aluminium or composite materials) and design modifications that could reduce weight and costs without compromising safety and performance.
6. **Evaluate Alternatives:** Tested the feasibility of using lighter materials and analysed their performance compared to the original design.
7. **Implement Changes:** Adopted a new design utilizing a combination of aluminium and reinforced plastics, significantly reducing weight while maintaining strength and safety.
8. **Review and Feedback:** Monitored the performance of the new car doors in production and gathered customer feedback regarding vehicle performance and aesthetics.

Results: The new design led to a **15% reduction in material costs** and a **10% weight reduction** in the vehicle, contributing to improved fuel efficiency. The initiative increased customer satisfaction and maintained safety standards.

Case Study 2: Consumer Electronics – Reducing Costs in a Smartphone

Background: A leading smartphone manufacturer wanted to reduce production costs for a popular model without sacrificing quality or performance. The company faced stiff competition and needed to maintain market pricing while improving profitability.

Value Analysis Process:

1. **Identify the Problem:** Rising production costs affecting profitability and competitive pricing.
2. **Gather Information:** Collected data on component costs, manufacturing processes, and market trends.
3. **Analyse Functions:** Identified essential functions such as communication, camera performance, and battery life, as well as secondary features like aesthetics and user interface.
4. **Assess Costs:** Evaluated the costs associated with each component (e.g., camera modules, screens, batteries) to understand where savings could be achieved.
5. **Generate Ideas:** Explored alternative suppliers, revised specifications for components (e.g., lower-cost camera sensors), and assessed the feasibility of modular designs for easier assembly and repair.
6. **Evaluate Alternatives:** Conducted performance tests on alternative components to ensure they met quality standards while offering cost savings.
7. **Implement Changes:** Integrated new suppliers and modified specifications for several components, leading to a more cost-effective production process.
8. **Review and Feedback:** Monitored production metrics and customer feedback on the new model's performance and quality post-launch.

Results: The changes resulted in a **20% reduction in production costs**, enabling the company to maintain competitive pricing without sacrificing quality. The revised model achieved strong sales, reinforcing the brand's position in the market.

Case Study 3: Aerospace Industry – Redesigning an Aircraft Component

Background:

An aerospace manufacturer faced increasing costs related to a specific aircraft component, the wing flap actuator, which was critical for performance but expensive to produce due to its complex design.

Value Analysis Process:

1. **Identify the Problem:** High production costs for the wing flap actuator affecting overall aircraft pricing and profitability.
2. **Gather Information:** Collected data on the existing actuator design, materials, manufacturing processes, and performance specifications.
3. **Analyse Functions:** Identified essential functions, including mechanical movement, reliability under stress, and lightweight design.
4. **Assess Costs:** Evaluated costs associated with materials (e.g., metals, plastics), machining processes, and assembly.
5. **Generate Ideas:** Explored alternative materials (e.g., lightweight composites) and simplified designs to reduce complexity while maintaining performance.
6. **Evaluate Alternatives:** Conducted tests to compare the performance of alternative designs and materials against the existing actuator specifications.
7. **Implement Changes:** Adopted a new design using a simplified mechanism and lightweight composites, leading to reduced manufacturing time and cost.
8. **Review and Feedback:** Monitored the performance of the new actuator during flight tests and gathered feedback from engineers on reliability and efficiency.

Results: The redesigned actuator resulted in a **25% reduction in production costs** and improved weight savings, enhancing overall aircraft performance.

Case Study 4: Healthcare Industry – Improving Medical Device Production

Background:

A medical device company sought to reduce production costs for a popular heart monitor while ensuring compliance with stringent regulatory standards.

Value Analysis Process:

1. **Identify the Problem:** Rising costs in the production of the heart monitor affecting pricing and profitability.
2. **Gather Information:** Collected data on production processes, components used, regulatory requirements, and market pricing.
3. **Analyse Functions:** Identified key functions such as accurate heart rate monitoring, user-friendliness, and data transmission capabilities.
4. **Assess Costs:** Analysed costs associated with each component, including sensors, software, and assembly processes.
5. **Generate Ideas:** Brainstormed alternative suppliers for components, potential design simplifications, and software optimizations.
6. **Evaluate Alternatives:** Tested new suppliers' components for reliability and compatibility while assessing design modifications.
7. **Implement Changes:** Integrated new suppliers and adopted a simplified design that retained functionality while streamlining production.
8. **Review and Feedback:** Monitored production metrics and collected user feedback post-implementation to evaluate performance and satisfaction.

Results: The changes led to a **30% reduction in production costs** and improved manufacturing efficiency, allowing the company to remain competitive in the healthcare market.

Case Study 5: Food Industry – Streamlining Snack Production

Background:

A snack food manufacturer wanted to reduce costs while improving the production process of its popular chip product, which was facing stiff competition.

Value Analysis Process:

1. **Identify the Problem:** High production costs and inefficiencies in the snack manufacturing process.
2. **Gather Information:** Collected data on current manufacturing processes, ingredient costs, and consumer preferences.
3. **Analyse Functions:** Identified essential functions such as flavour retention, texture, and shelf stability.
4. **Assess Costs:** Evaluated costs of raw materials (potatoes, seasonings), labour, and equipment used in production.
5. **Generate Ideas:** Explored alternative ingredients (e.g., using lower-cost seasonings) and process optimizations (e.g., batch sizes).
6. **Evaluate Alternatives:** Conducted taste tests and shelf-life studies to ensure that any changes would not compromise quality.
7. **Implement Changes:** Adopted new supplier agreements for ingredients and modified production processes to improve efficiency.
8. **Review and Feedback:** Gathered customer feedback and sales data post-implementation to assess product acceptance and profitability.

Results: The new approach resulted in a **15% reduction in production costs** while maintaining product quality, leading to an increase in market share.

Waste

Waste refers to any resource, material, time, or effort that is not utilized effectively or efficiently in the production process, resulting in unnecessary costs or loss of value. In the context of industrial management, waste encompasses various forms of inefficiencies that do not contribute to the creation of value for the customer. Waste can manifest in different ways, such as:

- **Excess inventory** that ties up capital and storage space.
- **Unproductive time** where resources or employees are idle.
- **Defective products** that require rework or disposal.
- **Inefficient processes** that slow down production without enhancing the product's value.

Types of Waste

1. **Overproduction:** Producing more than is needed or producing too early, leading to excess inventory.
2. **Waiting:** Time lost when processes are halted, waiting for materials, information, or equipment.
3. **Transportation:** Unnecessary movement of materials or products, which adds time and costs without adding value.
4. **Processing:** Inefficient processes that do not add value to the product or service, such as excessive steps or complicated workflows.
5. **Inventory:** Holding excess stock of materials or products, which can lead to waste due to spoilage, obsolescence, or increased holding costs.
6. **Motion:** Unnecessary movement of people or equipment, which does not add value, such as workers moving around to retrieve tools or materials.
7. **Defects:** Errors or defects in products that require rework or disposal, leading to wasted materials and labor.

8. **Unused Talent:** Underutilizing employees' skills and capabilities, which can lead to missed opportunities for improvement.

Sources of Waste

1. **Inefficient Processes:** Outdated or overly complicated workflows that slow down production.
2. **Poor Planning:** Inaccurate demand forecasting leading to overproduction or stockouts.
3. **Inadequate Training:** Employees not properly trained can result in mistakes and inefficiencies.
4. **Lack of Standardization:** Variability in processes can lead to inconsistencies and increased errors.
5. **Suboptimal Layout:** Poor facility layout can lead to unnecessary transportation and motion.

Ways to Reduce Waste

1. **Lean Manufacturing:** Implementing lean principles to eliminate waste by optimizing processes, improving workflow, and increasing efficiency.
2. **Just-In-Time (JIT) Production:** Producing only what is needed, when it is needed, to reduce inventory and overproduction.
3. **Value Stream Mapping:** Analysing the flow of materials and information to identify waste and areas for improvement.
4. **Employee Training and Engagement:** Training employees on best practices and encouraging them to contribute ideas for waste reduction.
5. **Process Standardization:** Creating standard operating procedures (SOPs) to reduce variability and improve efficiency.
6. **Continuous Improvement (Kaizen):** Fostering a culture of continuous improvement where small, incremental changes are made regularly.
7. **Investing in Technology:** Utilizing automation and advanced technologies to improve efficiency and reduce human error.
8. **Supplier Collaboration:** Working closely with suppliers to optimize inventory levels and reduce lead times.

Cost Control

Cost Control refers to the process of monitoring, managing, and regulating expenses within an organization to ensure that they align with budgeted amounts and financial objectives. The goal of cost control is to minimize unnecessary costs, optimize resource utilization, and improve overall profitability without compromising product quality or operational efficiency. **Cost Control Methods:**

1. **Budgeting:** Establishing budgets for various departments and projects to set spending limits and monitor financial performance.
2. **Variance Analysis:** Comparing actual costs to budgeted costs to identify discrepancies and investigate reasons for variances.
3. **Standard Costing:** Setting standard costs for products and services to identify inefficiencies and areas for improvement.
4. **Activity-Based Costing (ABC):** Allocating overhead costs based on actual activities and resource consumption to provide more accurate cost information.
5. **Cost-Benefit Analysis:** Evaluating the financial implications of decisions to ensure that benefits outweigh costs.
6. **Benchmarking:** Comparing performance metrics to industry standards or competitors to identify areas for improvement.

7. **Inventory Management:** Implementing effective inventory control measures to reduce holding costs and minimize waste.
8. **Supplier Negotiation:** Engaging in negotiations with suppliers to obtain better pricing or terms to reduce costs.

Important Guidelines for Cost Control

1. **Establish Clear Objectives:** Set specific, measurable objectives for cost control initiatives to provide direction and focus.
2. **Monitor Costs Regularly:** Regularly review financial performance and costs to identify trends and areas for improvement.
3. **Involve Employees:** Engage employees in cost control efforts by encouraging them to share ideas and suggestions.
4. **Use Technology:** Leverage technology and software solutions for accurate tracking and reporting of costs.
5. **Review Processes:** Continuously evaluate processes and systems to identify inefficiencies and areas where costs can be reduced.
6. **Promote a Cost-Conscious Culture:** Foster a culture within the organization that values cost control and encourages employees to be mindful of expenses.
7. **Communicate Transparently:** Keep lines of communication open regarding cost control initiatives, providing updates and soliciting feedback from stakeholders.
8. **Adapt and Adjust:** Be willing to adjust strategies and approaches based on changing circumstances or new information.